

O-RAN Open Fronthaul S-Plane — Software Testbed

Configuration, Setup and Standards Reference · Revised 2026-09-17 (supersedes the prior 81-run revision)

Document scope. Every hardware and software configuration used by the testbed, each traced to the standard or reference implementation that fixes it. No value is assumed — all were read back from the running system or from the cited standard.

Telecom profile. ITU-T G.8275.1 — PTP telecom profile for phase/time distribution with full timing support. Selected because O-RAN sync configs LLS-C1/C2/C3 distribute timing through the network, and because the real TIMESAFE hardware captures used for comparison run domain 24 (G.8275.1) on all 344,630 PTP packets across the six captures.

Frozen state. 18 artefacts SHA-256 hashed; `verify_frozen.sh` refuses to run if any changed and was passed before every admissible run.

Validation result (current). 132 admissible runs across 12 randomised replicates. Attack sensitivity 1.000 [0.940, 1.000] (n=60); benign specificity 0.800 [0.682, 0.882] (n=60) — 1.000 [0.926, 1.000] (n=48) once the single known schema limitation `B_bc_replacement` is excluded, which accounts for all 12 false positives; honest abstention 12/12; attack attribution 0.817 [0.701, 0.894]. Zero infrastructure failures.

Corrections in this revision (all pre-freeze, all verified). (1) Added `dataset_comparison G.8275.x` — it was missing, so `linuxptp` had been running the IEEE-1588 *default* BMCA rather than the G.8275.1 *alternate* BMCA. A capture cannot reveal this (the BMCA algorithm is not a wire field), so the earlier "verified by parsing its own capture" could not have caught it. (2) Removed `path_trace_enabled 1` — real TIMESAFE captures carry zero TLVs in all 344,630 PTP packets of the six captures, so leaving `PATH_TRACE` on makes its presence a perfect testbed-vs-TIMESAFE separator (a provenance leak). (3) Added a per-node config generator (`run/mkcfg.sh`); option scoping verified against `linuxptp v4.0 config.c`. (4) A 4th provenance leak was logged: `minor_version_ptp` (testbed 4.0 = 1, real = 0) — a compile-time constant, apparatus-only, excluded from any join.

1 Hardware / platform (software testbed)

Item	Value (read from the system)	How verified	Consequence
Kernel	Firecracker microVM	<code>uname -r</code>	Monolithic build
Privileges	uid 0, CAP_NET_ADMIN	<code>id -u;</code> <code>capsh --print</code>	Namespaces and bridges permitted
Loadable modules	None (no /lib/modules)	<code>modprobe</code> <code>ptp_mock</code> → not found	No per-node mock PHC
PTP hardware	None (no	<code>ls</code> <code>/dev/ptp*</code>	No hardware clock to discipline

clock	/dev/ptp*)		
veth timestamping	software only; PHC: none	ethtool -T	Microsecond noise floor vs a nanosecond target (measured: master offset median 0.9 μ s, p95 5.6 μ s; path delay median 7.2 μ s, p95 8.3 μ s (792 samples from the 132-run campaign's baseline runs))
Oscillator	One shared kernel clock	Architecture I	Clocks cannot drift apart; B1/B6 impossible

A second physical machine would add a second crystal (makes oscillator drift B6 real); a hardware-timestamping NIC + back-to-back cable drops the floor from μ s to tens of ns (makes A4/B5 measurable). Neither unlocks GNSS (A6/A7/B1) or SyncE (B4).

2 G.8275.1 profile configuration (Class A — exact)

Contents of `cfg/g82751.base`, applied to every node. Cross-checked line-by-line against `linuxptp v4.0's` own `configs/G.8275.1.cfg` and against the real TIMESAFE capture.

linuxptp key	Value	Standard / basis	Observed in real capture
dataset_comparison	G.8275.x	G.8275.1 alternate BMCA (ADDED this revision)	algorithm, not a wire field
domainNumber	24	G.8275.1 — telecom range 24–43	24 on all 344,630 PTP pkts ✓
logSyncInterval	-4 (16/s)	G.8275.1 mandated Sync rate	-4 on all Sync ✓
logAnnounceInterval	-3 (8/s)	G.8275.1 mandated Announce rate	-3 on all Announce ✓
logMinDelayReqInterval	-4 (16/s)	G.8275.1	—
announceReceiptTimeout	3	G.8275.1 (375 ms at 8/s)	—
priority1	128	G.8275.1 fixes it and removes it	128 on legitimate clocks

		from BMCA compare	✓
delay_mechanism	E2E	G.8275.1 requires end-to-end	Delay_Req/ Delay_Resp present ✓
network_transport	L2	G.8275.1 — Ethernet, no IP	EtherType 0x88F7 ✓
ptp_dst_mac	01:1 B:19: 00:0 0:00	G.8275.1 forwardable multicast	100% conformant ✓
path_trace_enabled	1	REMOVED this revision — PATH_TRACE provenance leak (real capture has 0 TLVs)	absent in real capture
G.8275.*.localPriority	128	G.8275.1 alternate-BMCA tiebreaker	—
maxStepsRemoved	255	IEEE 1588 / G.8275.1	—

Why this matters most. The project's earlier netem runs used domain 0, logSync -3, logAnnounce +1 (one Announce per 2 s vs the standard's 125 ms — a 16× mismatch). Because benign data came from that testbed and attack data from real G.8275.1 hardware, a classifier could have separated the two on cadence alone. Aligning the testbed to G.8275.1 and adding the alternate-BMCA comparison removes that confound.

3 Declared deviations from a production deployment

Setting	Why it differs	What it costs
free_running 1	All namespaces share one kernel clock; with the servo active they would fight over the same clock.	Offsets are measured and logged but not applied. Servo convergence dynamics absent.
time_stamping software	veth exposes no PHC (ethtool -T).	Microsecond floor. No G.8273.2 / G.8271.1 conformance claim.
clock_class_thresh old 248	An acceptance policy, not a quality claim; the lab has no PRTC/GPS so every node truthfully advertises clockClass 248.	No node given a fabricated clockClass 6; hierarchy uses priority2 (an operator field by design).

4 Topology and node roles

Two Ethernet bridges separated by a Boundary Clock (models O-RAN LLS-C2/C3).

```
[gma p2=100] [gmb p2=110] --- brUP --- [bc: v-bc-up | v-bc-dn] --- brDN ---  
[ru1] [ru2] [ru3]
```

Node	Role	priority2	Purpose
gma	Primary grandmaster (T-GM)	100 (best)	Wins BMCA normally
gmb	Provisioned backup GM	110	Enables B2 planned failover
bc	Boundary clock (T-BC)	120	stepsRemoved 0 → 1; enables A8 and B7
ru1/ ru2/ ru3	O-RU proxies (T-TSC)	128	Three clients = spatial axis

5 Scenario configurations (11 — randomised per replicate)

ID	Scenario	How produced on the wire	Class
—	baseline	All nodes nominal, GM-A → BC → 3 RUs	healthy
A1	Rogue grandmaster	ptp4l instance, randomised off-allow-list identity, randomised superior priority2 (1–90) and clockClass (6/7/13), self-announcing on RU segment	attack
A2	Sync spoof	scapy forged Sync (two-step flag set), randomised implausible originTimestamp and burst, unprovisioned source; never Announce	attack
A3	Replay	Live slice captured, retransmitted with tcpreplay (stale timestamps, reused sequenceId); randomised slice size and loop count	attack
A5	DoS / flood	scapy high-rate PTP flood (randomised burst/rate)	attack
A8	Rogue boundary	Two-port ptp4l (randomised priority2 105–118) slaving upstream, relaying downstream with	attack

	clock	stepsRemoved incremented	ck
B2	Planned GM failover	GM-A withdrawn, GM-B (on allow-list) elected, maintenance window open	benign
B_UNPLANNED	Unplanned GM crash (NEW)	GM-A killed, NO maintenance window; GM-B elected. Packet-indistinguishable from a suppression attack → expected verdict UNKNOWN (honest abstention)	benign
B_BCREPL	BC replacement (NEW)	In-path BC swapped for a provisioned replacement during maintenance. The replacement id is provisioned (expected_bc_identity_secondary), but the frozen rule reads only one BC identity → flagged A8; reported, not patched	benign
B3	PDV / congestion	netem normal-distribution delay (3/5/8 ms) ± jitter (2/3/4 ms) on RU links, randomised	benign
B7	Topology change	RU3 link bounced and re-homed during maintenance	benign

A1 vs B2 is the decisive pair. Both present a new grandmaster identity; they are separated only by provisioned context (allow-list + maintenance window). Without that context they are indistinguishable from the packets alone.

6 Faults deliberately out of scope (physical reason)

Fault	Why not producible/measurable here	What would be required
A6 GNSS spoof / A7 jam	No GNSS receiver in software; faking a status enum is inventing evidence	GNSS simulator + receiver + RF-shielded chamber
B1 GNSS holdover	No reference to lose, no characterised oscillator	GNSS timing receiver + independent oscillator
B4 SyncE / EEC	Physical-layer function; veth has no PHY	SyncE-capable NIC + switch, sync4l
B6 Oscillator drift	All namespaces share one crystal	A second physical machine
A4 Delay	Injectable, but a passive capture	Live per-client pmc telemetry or a

attack	yields only t1,t4 (t2,t3 never on the wire)	redundant path; HW timestamping for ns claims
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7 Validity controls

Control	How enforced
No tunable parameters	Every constant in the decision rule is a standard, a provisioned fact, or a structural definition. The one self-chosen definition is the 1% transient filter (<code>TRANSIENT_FRACTION = 0.01</code>). Its standard replacement (IEEE 1588 §9.3.2.4, <code>FOREIGN_MASTER_THRESHOLD = 2</code> within 4 Announce intervals) has been identified but not adopted, because adopting it needs a re-freeze
Pre-registration	<code>freeze.sh</code> SHA-256 hashes 18 artefacts; <code>verify_frozen.sh</code> passed before every admissible replicate
Development /validation split	Reps that exposed defects are development data and are excluded; only post-freeze reps (30–41) are scored
Randomisation	Per-rep attacker identities, priorities, timings and burst sizes (<code>run/randparams.py</code> , seeded by rep) — runs are genuinely independent, so CIs are honest not optimistic
Session-disjoint splitting	Evaluation by run, never by window
Fail-closed	Missing/unreadable evidence → UNKNOWN, never BENIGN; infrastructure failures retried once then discarded with disclosure
Leak audit	<code>run/leak_check.py</code> checks every field for testbed-vs-real separability; 4 leaks found: 3 eliminated by the G.8275.1 testbed (16× cadence confound, synthetic <code>offsetScaledLogVariance</code> 4096-vs-65535, <code>PATH_TRACE</code>), and 1 that cannot be fixed by configuration (<code>minor_version_ptp</code>) excluded from any testbed+real join

8 Validation result

Metric	Value	95% Wilson CI	n
Sensitivity (attack recall)	1.000	[0.940, 1.000]	60
Specificity — all benign runs (TN 48, FP 12)	0.800	[0.682, 0.882]	60
Specificity — excluding the known <code>B_bc_replacement</code> schema limitation	1.000	[0.926, 1.000]	48
Abstention correctness (ambiguous benign →	1.000	[0.757, 1.000]	12

UNKNOWN)

Attack attribution (correct fault id)	0.817	[0.701, 0.894]	60
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Binary attack detection 60/60 (TP=60, FN=0). All 12 specificity false positives are the single B_bc_replacement schema limitation. Attribution confusion A1 → A3 (×4) and A8 → A3 (×7) is the documented re-parenting effect (rogue master/BC insertion causes real sequenceId restarts, caught by the A3 clause first); it does not change the binary ATTACK verdict. Cross-domain: the frozen rule, applied unchanged to six real TIMESAFE hardware captures it never saw, classified **6 of 6 as ATTACK**; **5 of 6** on standards/protocol rules alone (3 by Announce-field legality, 2 by IEEE 1588 cl.11.3 sequenceId monotonicity). The remaining one is a healthy-looking spoof with fully conformant fields and needs the provisioned allow-list.

Artefacts: results_2026-09-17/ (EVALUATION.json, TIMESAFE_CROSSVAL.json, FROZEN.json, SESSION_FINDINGS.md, g87251_testbed_v2.tgz, full 132-run raw archive). Prior 81-run revision retained in _pre_2026-09-17_doc_update/.